

From Logical to Concrete: OpenSCENARIO 2.0–Driven Scenario Extraction from Naturalistic Driving Data

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Abstract—Scenario-based virtual validation relies on logical scenario specifications that define classes of traffic situations through parameterized descriptions. These logical scenarios are typically sampled to generate finite sets of executable, concrete simulation instances. However, existing approaches typically do not directly leverage such formal logical specifications to extract corresponding concrete instances from naturalistic driving datasets (NDD). This paper presents a formal scenario extraction pipeline that uses OpenSCENARIO 2.0 logical descriptions as constraint-generating specifications for data-driven scenario identification. From each logical scenario, compositional and temporally anchored constraints are derived and evaluated over reconstructed road-aligned actor traces. To enable scalable and semantically consistent filtering, the method uses graph-based lane topology reconstruction from NDD map primitives to aggregate road segments with stable lateral adjacency relations, and establishes a deterministic OpenDRIVE-compatible lane indexing scheme for relational reasoning. Concrete scenario instances are identified by evaluating constraints derived from OpenSCENARIO 2.0 logical scenarios over discretely sampled naturalistic driving trajectories. The approach is evaluated on the Waymo Open Motion Dataset for three ADAS-relevant scenario types. The results demonstrate scalable search-space reduction, deterministic semantic constraint enforcement, and realistic parameter domains derived directly from real-world data.

I. INTRODUCTION

Increasing levels of vehicle automation and autonomy place growing demands on systematic safety assurance and regulatory compliance throughout the development process. Scenario-based testing has therefore become a central validation paradigm, both in virtual environments (X-in-the-Loop setups) and on proving grounds, enabling structured evaluation within a defined operational design domain (ODD) and reproducible coverage analysis [1], [2]. However, while scenario-based methodologies are well established in controlled validation environments, the systematic identification and coverage analysis of formally defined scenarios within large-scale naturalistic driving datasets remains largely unresolved. Bridging this gap requires a method that connects formal scenario specifications with raw trajectory data in a deterministic and reproducible manner.

In this context, large-scale naturalistic driving datasets (NDD), such as the Waymo Open Motion Dataset [3], inD [4], and INTERACTION [5], provide extensive recordings of real-world traffic interactions. These datasets represent a valuable empirical basis for deriving realistic sce-

nario parameters and observing naturally occurring traffic behavior. However, they are typically unordered (not labeled or indexed according to scenario definitions) collections of trajectories and map primitives. In addition, they lack explicit road-level topology and stable relational lane information required to evaluate constraints from structured scenarios.

As a result, identifying safety-relevant scenarios in NDD often relies on heuristic tagging, threshold-based filtering, or trained classifiers that are dataset-specific and not directly grounded in formal scenario semantics. This limits reproducibility, comparability, and alignment with standardized scenario description frameworks.

This raises the following research question: *How can formally defined safety-critical traffic scenarios, specified in ASAM OpenSCENARIO 2.0, be automatically and reproducibly identified in unordered naturalistic driving data?*

To address this question, we propose a specification-driven extraction pipeline that establishes a deterministic mapping from logical OpenSCENARIO 2.0 scenario definitions to concrete subtraces in naturalistic driving datasets. The contributions of this paper are:

- A structured method for identifying and extracting concrete scenario instances from unordered NDD based on OpenSCENARIO 2.0 descriptions.
- A map-based search-space reduction that reconstructs missing lane topology and aggregates topology-invariant road segments, enabling scalable scenario mining in large-scale datasets.
- A transformation of reconstructed road segments into an OpenDRIVE-compatible representation with deterministic lane indexing for relational reasoning.
- An evaluation of the Waymo Open Motion Dataset demonstrating the extraction of three ADAS-relevant scenario types under strict semantic constraint enforcement.

The remainder of this paper is structured as follows: Section II reviews related work. Section III presents the formal extraction pipeline. Section IV describes the experimental evaluation, followed by discussion and conclusions in Section V.

II. RELATED WORK

Scenario-based testing and scenario extraction for automated driving have been widely studied in recent years, and multiple surveys provide comprehensive overviews of the field [6]–[10]. These works examine scenario generation, critical scenario identification, and data-driven valida-

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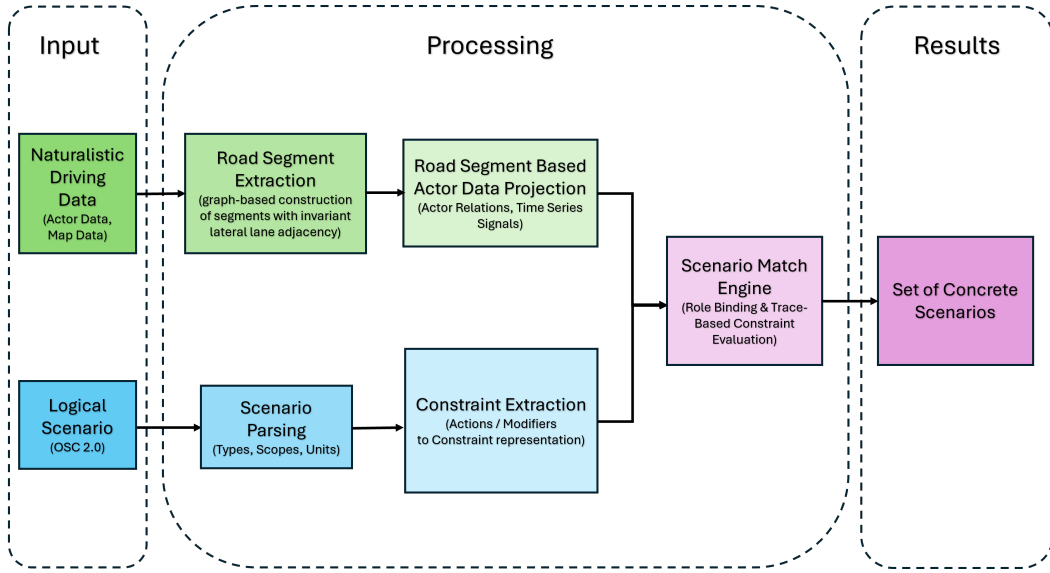


Fig. 1. Overview of the proposed formal scenario extraction pipeline.

tion methodologies across knowledge-based and data-driven paradigms.

In the context of large naturalistic driving datasets (NDD), scenario extraction approaches can broadly be grouped into (i) criticality- and rarity-driven mining, (ii) query- or language-based retrieval, (iii) tagging- and classification-based identification, and (iv) formal interaction mining for dataset construction. In contrast to these approaches, the present work performs a deterministic, specification-driven mapping from OpenSCENARIO 2.0 logical definitions to concrete substraces in unordered NDD.

A. Corner-Case Mining and Criticality-Driven Extraction

Corner-case mining approaches aim to identify rare but safety-critical situations by combining risk metrics with statistical rarity measures. Zhao *et al.* [11] construct a corner-case dataset from multi-source naturalistic driving data and introduce a quantitative marginality metric combining a risk-based term (e.g., minimal time-to-collision TTC) with a rarity term derived from maneuver and scenario element frequencies.

Such approaches emphasize coverage of rare and high-risk events; however, scenario identification is driven by criticality metrics rather than by direct evaluation of formally specified logical scenario definitions. The extraction logic is therefore metric-based rather than specification-driven.

B. LLM- and Query-Based Scenario Retrieval

Recent work explores natural-language and semantic query frameworks as interfaces for scenario search in trajectory datasets. Chat2Scenario [12] maps textual descriptions (e.g., cut-in events) to structured activity and positional classifications and retrieves matching trajectories from datasets such as highD [13]. The retrieved scenarios can subsequently be exported to OpenSCENARIO and simulation environments.

While such approaches improve usability and flexibility, scenario matching is typically based on dataset-specific activity labels and heuristic filtering. The logical structure and

compositional semantics of OpenSCENARIO 2.0 are not directly evaluated over the raw trajectory data.

C. Tagging, Rule-Based, and Learning-Based Identification

Tag-based scenario extraction frameworks describe traffic situations as combinations of actor activities, infrastructure interactions, and relational conditions. Guo *et al.* [14] introduce a multi-stage pipeline involving trajectory preprocessing, semantic tagging, and scenario retrieval via tag matching, applied to large real-world datasets. Similarly, Kayatas and Bestle [15] compare rule-based lane-change detection with a learning-based time-series classification approach trained on synthetic maneuver data, demonstrating improved detection performance of the data-driven model.

These methods effectively detect predefined maneuver classes but typically rely on handcrafted tags, dataset-specific thresholds, or trained classifiers. They do not evaluate compositional logical scenario definitions expressed in OpenSCENARIO 2.0, nor do they reconstruct missing road topology to enable formal relational lane reasoning.

D. Formal Interaction Mining and Dataset Construction

Recent dataset-oriented work has employed formal interaction definitions to extract structured multi-agent events from large-scale naturalistic driving recordings. The InterHub dataset [16] reorganizes multiple public NDD sources into a unified representation and uses formally defined spatio-temporal interaction criteria to mine dense multi-agent interaction segments. Extracted events are characterized by interaction intensity, agent count, and conflict measures, enabling reproducible dataset construction.

While InterHub introduces formalized interaction extraction, its objective is to curate and annotate interaction datasets rather than to evaluate compositional scenario specifications. In contrast, the present work performs specification-driven extraction based on OpenSCENARIO 2.0 semantics, combining topology reconstruction,

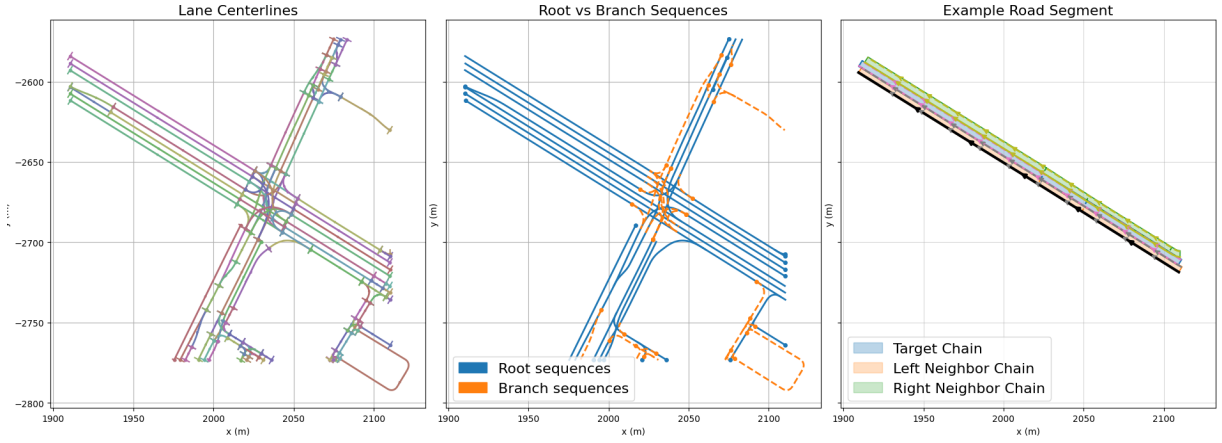


Fig. 2. Overview of the road segment extraction and normalization steps.

segment-relative normalization, and deterministic trace evaluation to map logical scenario definitions to concrete instances.

E. Positioning of This Work

In contrast to the approaches reviewed above, this paper contributes a formally grounded and reproducible mapping between formal OpenSCENARIO 2.0 scenario semantics and observed trajectory behavior in naturalistic driving data (NDD) without predefined scenario structure or labeling.

The approach (i) reconstructs missing lane topology to enable stable relational reasoning, (ii) constructs OpenDRIVE-compatible road segments from reconstructed lane topology, such that lane adjacency remains invariant within each segment, enabling deterministic lane indexing, and (iii) evaluates compositional and temporal OpenSCENARIO 2.0 semantics over observed traffic trajectories.

This establishes a direct correspondence between formal scenario specifications and raw trajectory data without relying on heuristic tagging or learned classifiers.

III. FORMAL SCENARIO EXTRACTION PIPELINE

This section presents the formal extraction pipeline that maps logical OpenSCENARIO 2.0 specifications onto naturalistic driving data, thereby identifying scenario instances within the observed trajectories.

The method consists of two parallel modeling streams that converge in a deterministic semantic matching procedure:

- **Data stream:** Road- and actor-level representation derived from naturalistic driving data (NDD).
- **Scenario stream:** Constraint extraction from logical OpenSCENARIO 2.0 specifications.

The method constructs a road-aligned representation from raw map and actor data, extracts formal constraints from the OpenSCENARIO 2.0 specification, and evaluates their associated predicates over the dataset-derived actor traces.. An overview of the complete pipeline is shown in Fig. 1.

Logical OpenSCENARIO 2.0 specifications are interpreted as finite sets of formal constraints. Each constraint is operationalized as a Boolean predicate evaluated over segment-relative system states.

A. Data Stream: Road and Actor Modeling

1) *Road Segment Extraction:* The Waymo Open Motion Dataset (WOMD) map representation provides individual lane elements defined by sampled 3D centerline points and lane attributes (type, width). An example WOMD scene is shown in the left image of Fig. 2, where different lane elements are colored by ID. However, the dataset does not provide an explicit successor graph, lateral neighbor relations, or a structured road-level topology. In particular, predecessor/successor connectivity and left/right adjacency between lane elements are not directly available.

This limitation prevents direct evaluation of relational scenario constraints such as longitudinal ordering within a shared lane or relative lane positioning between vehicles (e.g., vehicle A located 100m ahead of vehicle B in a lane to the right). Such predicates require both longitudinal connectivity and stable lateral adjacency relations.

To reconstruct a consistent lane topology suitable for relational reasoning, the centerlines of lane elements (given as 3D point sequences) are modeled as nodes V of a directed graph

$$G = (V, E), \quad (1)$$

where edges E represent geometric continuity inferred via KD-tree proximity between lane element endpoints.

Nodes with zero in-degree define *root nodes*. Starting from each root lane element, a root sequence is constructed by iteratively following successor elements. If multiple successors exist, the geometrically best-aligned successor (minimal orientation deviation) continues the root sequence, while alternative successors initialize branch sequences. In this work, road construction is restricted to root sequences to ensure linear ordering and avoid ambiguity at merge or split regions; branch sequences are discarded. The middle image of Fig. 2 illustrates this separation.

Each lane element in a root sequence is subdivided into fixed-length 5 m lane segments to enable localized topology analysis.

An STRtree spatial index is constructed over all lane segments. For each segment:

- A lateral buffer polygon is generated using WOMB lane width information,
- Overlapping candidate neighbor segments are retrieved via STRtree queries,
- Same-lane longitudinal continuity is excluded,
- Directional consistency is verified,
- Left/right relations are determined using orientation-based tests.

Consecutive 5 m lane segments along a root sequence are aggregated into higher-level structures.

Definition (Topology-Invariant Road Segment). A topology-invariant road segment (TIRS) r_i is defined as a maximal contiguous interval of lane segments along a root sequence for which lateral adjacency relations remain constant over the entire interval.

Each TIRS r_i is characterized by:

- a set of participating WOMB lane IDs $L(r_i)$,
- a contiguous index interval of 5 m lane segments,
- a constant left/right adjacency graph over that interval.

The output of this stage is a set of topology-invariant road segments $\{r_i\}$ with invariant longitudinal and lateral topology. These segments form the structural basis for actor data modeling and subsequent scenario matching.

2) *Road Segment Based Actor Data Projection:* Each topology-invariant road segment (TIRS) r_i is transformed into an OpenDRIVE-compatible representation [17] to establish a standardized longitudinal and lateral reference frame.

For each r_i , a reference line is constructed from the ordered lane centerlines of the segment, yielding a continuous arc-length coordinate $s \in [0, |r_i|]$ along the reference line of the segment. This defines a local coordinate system w.r.t. r_i .

We define a segment-local projection operator

$$\pi_{r_i} : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \quad x_a(t) \mapsto (s_a^{r_i}(t), t_a^{r_i}(t)) \quad (2)$$

which maps a global actor position to longitudinal arc-length s and signed lateral offset t within r_i .

We denote the segment-restricted actor state by

$$x_a^{r_i}(t) := \pi_{r_i}(x_a(t)) = (s_a^{r_i}(t), t_a^{r_i}(t)). \quad (3)$$

where $x_a(t)$ is the global position of actor a at time t and $x_a^{r_i}(t)$ is its representation in the local coordinate system induced by r_i .

A segment-specific lane indexing function

$$\mu_{r_i} : L(r_i) \rightarrow \mathbb{Z} \quad (4)$$

assigns OpenDRIVE-compatible lane indices to lane elements based on their lateral ordering with respect to the reference line of r_i .

This enables evaluation of relational predicates such as `same_lane`, `left_of`, and `right_of` via integer comparisons of lane indices within a fixed TIRS.

The associated lane index is given by:

$$\ell_{\text{XODR},a}(t) = \mu_{r_i}(\text{lane}(x_a^{r_i}(t))). \quad (5)$$

Longitudinal and lateral velocities are approximated via finite differences:

$$\dot{x}_a^{r_i}(t) \approx \frac{x_a^{r_i}(t + \Delta t) - x_a^{r_i}(t)}{\Delta t}. \quad (6)$$

$$\dot{x}_a^{r_i}(t) = (\dot{s}_a^{r_i}(t), \dot{t}_a^{r_i}(t)). \quad (7)$$

The resulting segment-relative actor state representation within a TIRS is with associated segment r_i :

$$(x_a^{r_i}(t), \dot{s}_a^{r_i}(t), \dot{t}_a^{r_i}(t), \ell_{\text{XODR},a}(t)). \quad (8)$$

This formulation ensures that all relational reasoning is performed within a topology-consistent spatial domain defined by r_i , enabling deterministic evaluation of scenario constraints. All state components are implicitly conditioned on the TIRS r_i through the projection π_{r_i} and the lane indexing function μ_{r_i} .

B. Scenario Stream: Constraint Extraction

1) *Scenario Parsing:* Each logical OpenSCENARIO 2.0 specification is parsed and semantically validated. The compositional structure (serial and parallel constructs), actions, and modifiers are reconstructed and flattened into atomic executable elements.

The parser resolves:

- `serial` compositions representing ordered scenario phases,
- `parallel` compositions representing concurrently valid elements,
- actions and associated modifiers with temporal anchors.

2) *Constraint Extraction and Formalization:* In the following, a *constraint* denotes a formal semantic requirement derived from an OpenSCENARIO 2.0 specification, while a *predicate* denotes its executable Boolean interpretation over segment-relative actor traces. Constraint satisfaction is therefore determined through predicate evaluation over discrete subtraces.

Each atomic action and modifier is mapped to a formal constraint over segment-relative actor-state components such as the longitudinal coordinate $s_a(t)$, lane index $\ell_{\text{XODR},a}(t)$, and inter-actor relations such as `behind_of`, `in_front_of`, `same_lane`, `left_of`, and `right_of`.

Temporal anchors determine whether the resulting constraint belongs to the start, during, or end phase of the scenario.

From this mapping, a finite set of formal constraints is generated and categorized as Start-, During-, and End-constraints.

The scenario stream therefore produces a finite set of formal constraints together with associated Boolean predicates evaluated over dataset-derived actor traces.

The data stream yields segment-relative actor traces, while the scenario stream yields a finite set of formal constraints. Both representations are combined in the scenario match engine, which evaluates discrete trace satisfaction under role bindings.

C. Scenario Match Engine

1) *Discrete Trace Interpretation:* OpenSCENARIO 2.0 defines a scenario trace as a continuous-time sequence of system states [18]. Formally, a trace is defined as

$$\pi : \mathbb{R}_{\geq 0} \rightarrow S, \quad (9)$$

where S denotes the set of system states.

OpenSCENARIO 2.0 defines a trace as a continuous-time system state evolution. In this work, we interpret this semantics over discretely sampled system states obtained from the dataset.

A discrete subtrace

$$\tau[t_1, t_2] \quad (10)$$

is therefore represented as a finite sequence of sampled system states.

At each discrete time step t , we define a discrete instantiation of the continuous system state:

$$s_{r_i}(t) = \{x_a^{r_i}(t) \mid a \in \mathcal{A}\}, \quad (11)$$

where $x_a^{r_i}(t)$ denotes the segment-relative state of actor a at time t .

TABLE I
DERIVED PARAMETER CONSTRAINTS FROM THE OPENSCENARIO SPECIFICATION.

Parameter	Range	Temporal Anchor
Maneuver duration	[4, 9] s	during (window)
Ego speed	[30, 120] km/h	during
NPC speed	[30, 120] km/h	during
NPC distance ahead	[5, 50] m	start
NPC distance ahead	[5, 50] m	end

2) *Role Binding:* A logical scenario specification in OpenSCENARIO 2.0 defines a finite set of actor-typed scenario parameters. We interpret these parameters as roles and denote the resulting role set by $\mathcal{R}$.

A role binding is obtained by selecting one compatible scene actor for each role, thereby inducing all valid role-to-actor assignments in the scene, with the constraint that each scene actor is assigned to at most one role within a binding. Formally, a role binding is an injective mapping

$$\beta : \mathcal{R} \rightarrow \mathcal{A}, \quad (12)$$

where $\mathcal{A}$ denotes the set of scene actors.

3) *Predicate Evaluation via SEDs:* Given a role binding and a discrete trace segment induced by a topology-invariant road segment r_i , all atomic constraints are first evaluated over the full segment-conditioned trace $x_a^{r_i}(t)$ for $t = 1, \dots, T$.

This yields, for each atomic constraint, a Boolean time series over the complete evaluation horizon:

$$B_p \in \{0, 1\}^T. \quad (13)$$

These predicate evaluations are computed once per segment and reused for all candidate temporal intervals.

Candidate behavior-block intervals $\tau[t_s, t_e]$ are then applied as temporal restrictions over these precomputed

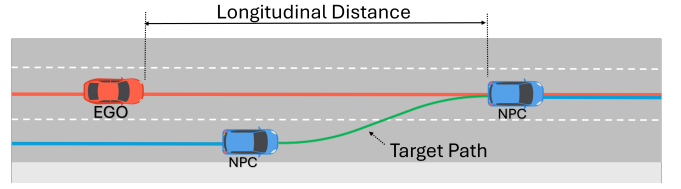


Fig. 3. Conceptual flow of a Cut-In scenario. The NPC starts in the adjacent lane, performs a lane change into the ego lane within a bounded time interval, and maintains a constrained longitudinal gap before and after the maneuver.

Boolean time series, selecting the relevant sub-sequences corresponding to start, during, and end phases.

We group constraints according to their temporal phase (start, during, end) and define a phase-wise evaluation structure. Each phase yields a Boolean time series over the full segment-conditioned trace.

Formally, a Start–During–End (SED) evaluation is defined as:

$$\text{SED} = (B_s, B_d, B_e), \quad (14)$$

where each $B_k \in \{0, 1\}^T$ is obtained by pointwise conjunction of all atomic predicate evaluations associated with phase k over the full segment-conditioned trace.

Behavior-block constraints are enforced by restricting the phase-wise Boolean time series to the corresponding temporal window $\tau[t_s, t_e]$, and evaluating start-, during-, and end-conditions directly on these restricted sub-sequences.

4) *Constraint Satisfaction:* A candidate subtrace satisfies the scenario specification if all phase-wise constraint groups satisfy the compositional and temporal requirements of the original OpenSCENARIO 2.0 specification.

Evaluation proceeds by (i) computing SEDs for each behavior block over its associated interval, and (ii) combining block-wise results according to serial and parallel composition semantics.

This yields a deterministic discrete trace-satisfaction procedure consistent with OpenSCENARIO 2.0 semantics.

D. Illustrative Example: Cut-In Scenario

To illustrate the complete extraction pipeline, we consider a representative Cut-In scenario. Fig. 3 shows the conceptual evolution of the maneuver. Initially, the NPC is located in the neighboring lane ahead of the ego vehicle. Within a specified duration window, the NPC performs a lane change into the ego lane. Both at the start and at the end of the maneuver, bounded longitudinal distances between NPC and ego vehicle are required.

Formal Scenario Specification: The logical scenario is defined using ASAM OpenSCENARIO 2.0. Listing 1 shows the complete specification. The specification encodes:

- A serial composition containing a parallel maneuver phase,
- A bounded maneuver duration of [4, 9] seconds,
- Speed constraints for both actors,
- Initial and terminal longitudinal gap constraints,

- A lane change action with explicit start and end lane relations.

Constraint Derivation: From the formal specification, atomic constraints are extracted during scenario parsing and grouped according to their temporal anchors (at start, during, at end).

```
import standard.osc

scenario top:
  path: path
  path.min_lanes(2)
  ego_vehicle: vehicle
  npc: vehicle

do serial:
  change_lane: parallel(duration: [4s..9s]):
    ego_vehicle.drive(path) with:
      speed([30kph..120kph])
      lane(-1, at: start)
      keep_lane()
  npc.drive(path) with:
    speed([30kph..120kph])
    position([5m..50m], ahead_of:
ego_vehicle, at: start)
    lane(-2, at: start)
    change_lane(1, side: left)
    lane(same_as: ego_vehicle, at: end)
    position([5m..50m], ahead_of:
ego_vehicle, at: end)
```

Listing 1. Cut-In scenario in OpenSCENARIO 2.0.

Table I summarizes the derived parameter constraints.

Each extracted constraint is translated into a Boolean predicate over components of the segment-relative actor state introduced in Section III. For each candidate role binding and time window, the corresponding predicate evaluations generate SED evaluations over the discrete subtrace, which are used to determine constraint satisfaction for:

- Start constraints (initial lane and distance relations),
- During constraints (speed bounds and duration),
- End constraints (final lane alignment and distance bounds).

Specifically:

- The predicates corresponding to the speed constraints are evaluated using the longitudinal velocity $\dot{s}_a^{r_i}(t)$ derived from projected trajectories.
- Longitudinal distance constraints $\text{position}([5m..50m], \text{ahead_of})$ are evaluated as

$$s_{\text{npc}}^{r_i}(t) - s_{\text{ego}}^{r_i}(t) \in [5, 50] \text{ m.} \quad (15)$$

- Lane predicates such as `lane(-1, at: start)` and `lane(same_as: ego_vehicle, at: end)` are evaluated via the segment-specific OpenDRIVE lane indices $\ell_{\text{XODR},a}(t)$.
- The modifier `keep_lane()` is evaluated by verifying that the OpenDRIVE lane index $\ell_{\text{XODR},a}(t)$ remains constant over the corresponding time interval.
- The modifier `change_lane(1, side: left)` is evaluated by detecting a discrete transition of the OpenDRIVE lane index $\ell_{\text{XODR},a}(t)$ by +1 in accordance with

the established segment-specific lane ordering. Additionally, monotonic lateral motion in $t_a^{r_i}(t)$ is verified during the maneuver interval

- The duration constraint $[4s..9s]$ is checked over the discrete subtrace length induced by the dataset sampling.

Each predicate evaluation yields a Boolean SED evaluation over the discrete subtrace. Start, during, and end constraints are evaluated according to their temporal anchors and combined using the serial-parallel composition semantics of OpenSCENARIO 2.0.

A candidate subtrace is accepted as a Cut-In instance if and only if all grouped predicate evaluations satisfy the compositional semantics of the OpenSCENARIO specification.

E. Search-Space Reduction Characteristics

As described in Section III-A, road segments with invariant lateral adjacency were constructed from the WOMD lane elements. Across the complete dataset, the procedure generated 1,358,623 two-lane and 674,802 three-lane road segments. To estimate coverage, 50 randomly selected scenes from each of the 1,000 TFRecord files were analyzed. The total considered drivable lane length amounts to approximately 15.9 million meters. Of this, approximately 9.45 million meters lie within the constructed road segments, corresponding to a coverage of 58.7% of the relevant lane lengths.

This demonstrates that a substantial portion of the structured roadway network can be represented by segments with stable relational topology, enabling consistent scenario evaluation.

F. Extracted Scenario Statistics

Three OpenSCENARIO 2.0 scenario types were evaluated:

- Cut-In,
- Car-to-Car Rear Braking (CCRB),
- Car-to-Pedestrian Nearside Adult (CPNA).

Table II summarizes the number of detected concrete scenario instances in the WOMD.

TABLE II
NUMBER OF EXTRACTED CONCRETE SCENARIO INSTANCES IN WOMD.

Scenario Type	Instances
Cut-In	9,814
CCRB	10,244
CPNA	4,456

To quantify scenario binding occurrence probability, the proportion of valid role bindings for which at least one admissible time window satisfies the scenario specification was computed:

$$P_{\text{binding}} = \frac{\text{bindings with hits}}{\text{enumerated valid bindings}}. \quad (16)$$

Table III reports the resulting values.

These results indicate that all three formally specified scenario types occur naturally in the dataset, albeit with different relative frequencies.

TABLE III
SCENARIO TYPICALITY P_{BINDING} .

Scenario	P_{binding}
Cut-In	1.7×10^{-4}
CCRB	1.8×10^{-4}
CPNA	4.39×10^{-3}

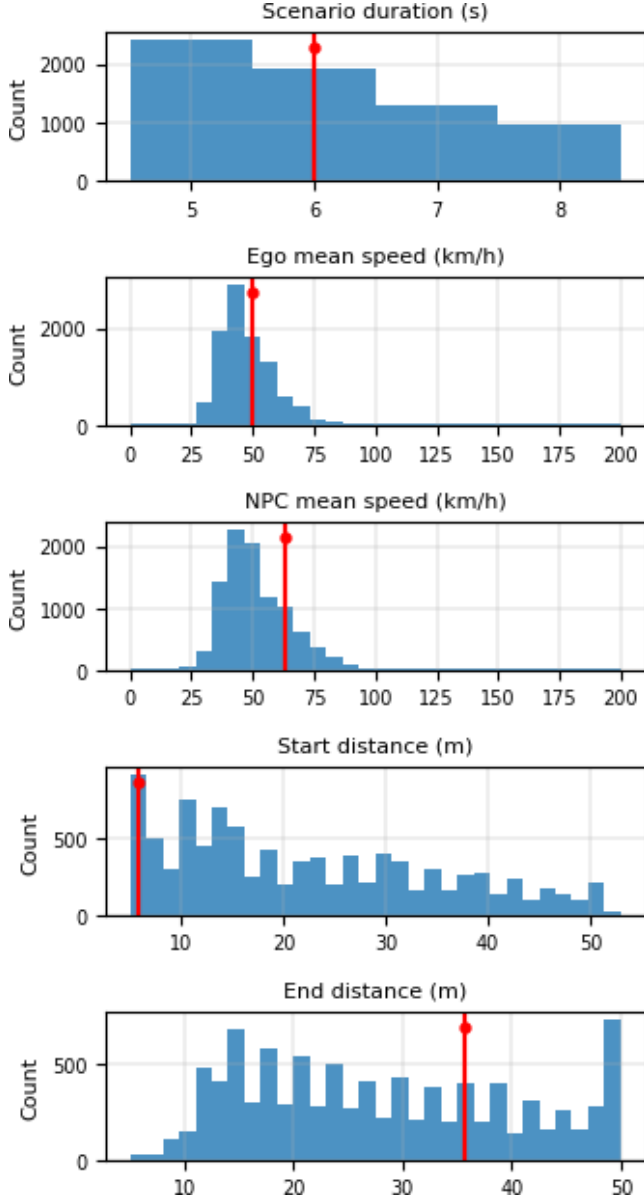


Fig. 4. Parameter Histograms for Cut-In Scenario. The red vertical lines mark the parameter values of the example Cut-In instance shown Fig. ??.

G. Parameter Distributions and Constraint Consistency

For all extracted scenario instances, parameter values corresponding to start, during, and end constraints were recorded and aggregated. Fig. 4 shows histograms for the Cut-In scenario parameters. The red vertical lines mark the parameter values of the example Cut-In instance shown Fig. 5.

Across all scenario types:

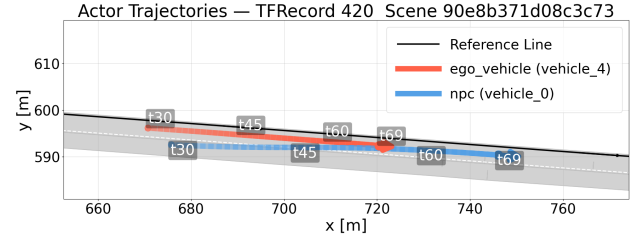


Fig. 5. Concrete Instance of Cut-In Scenario found in the WOMB dataset.

- Observed parameter values lie within the formally specified constraint intervals,
- Start and end constraints are satisfied at the corresponding subtrace boundaries,
- During constraints hold across the full accepted interval.

No violation of parameter bounds was observed among accepted subtraces. This confirms that the extraction procedure enforces the formal OpenSCENARIO 2.0 constraints under the discrete trace interpretation.

H. Cross-Scenario Interpretation

The considered scenario types differ significantly in both typicality and parameter distributions. While Cut-In and CCRb exhibit low binding probabilities, they nevertheless occur consistently across the dataset. The CPNA scenario shows higher relative typicality, reflecting the broader constraint definitions and frequent urban interactions.

Histogram-based parameter analysis reveals both common operating regions and rare boundary cases within each scenario type. These empirical distributions provide realistic parameter domains for scenario-based ADAS validation. Rather than defining artificial parameter combinations, the extracted distributions enable data-driven sampling of plausible test scenarios.

It should be noted that the reported distributions are dataset-dependent and reflect only the traffic situations present in WOMB under the applied scenario definitions. They do not represent general traffic occurrence probabilities.

IV. DISCUSSION

A. Deterministic Semantics and Evaluation Implications

The proposed approach performs semantics-driven scenario extraction by checking discrete trace satisfaction under a role binding, rather than learning a classifier or scoring similarity to a template. Consequently, standard predictive metrics such as precision/recall are not directly applicable without an external ground-truth labeling of scenario occurrences. Instead, evaluation in this work focuses on (i) scalability via map-based search-space reduction, (ii) occurrence statistics via binding typicality, and (iii) semantic consistency via parameter distributions that remain within the formally specified constraint bounds. The experiments validate deterministic constraint satisfaction, search-space reduction, and the existence of semantically admissible real-world instances; they do not claim exhaustive ground-truth detection performance.

B. Limitations

The current pipeline restricts relational reasoning to road segments constructed from root lane sequences. This provides stable neighborhood topology but excludes more complex infrastructure such as merges, splits, and intersections. Moreover, OpenSCENARIO 2.0 trace semantics are evaluated on discretely sampled trajectories; short-lived events below the sampling resolution or annotation noise may therefore affect boundary placement of accepted subtraces. Finally, the reported distributions and typicality values are dataset- and definition-dependent: they reflect WOMB traffic coverage and the chosen OpenSCENARIO 2.0 specifications rather than general traffic occurrence probabilities.

C. Practical Implications

Despite these limitations, the pipeline provides a direct semantic bridge from formal OpenSCENARIO 2.0 specifications to simulation-ready scenario instances. The OpenDRIVE-normalized road representation enables consistent relational lane predicates, while the extracted parameter distributions support data-driven parametrization of test scenarios (e.g., sampling around empirical modes for nominal testing and using observed boundary cases for stress testing). This facilitates reproducible scenario mining and reuse across virtual validation toolchains.

V. CONCLUSION

This paper addressed the problem of automatically identifying formally defined traffic scenarios in unordered naturalistic driving data. While scenario-based validation relies on precise logical specifications, large-scale trajectory datasets lack explicit topological structure and a direct linkage to formal scenario semantics.

We presented a formal extraction pipeline that reconstructs lane-level road topology from naturalistic driving data, derives road-aligned actor representations, and evaluates OpenSCENARIO 2.0 specifications through deterministic constraint checking over subtraces of system states. The approach establishes a reproducible mapping from logical scenario definitions to concrete subtraces in naturalistic driving data.

Application to the Waymo Open Motion Dataset demonstrated scalable search-space reduction and successful extraction of three ADAS-relevant scenario types under strict constraint enforcement. The resulting parameter distributions reflect realistic operating domains derived directly from real-world data.

The presented method is currently limited to road segments derived from root lane sequences. Future work will extend the approach to more complex infrastructure elements such as intersections and merge regions, and investigate integration of criticality-based prioritization for scenario ranking.

Overall, the proposed pipeline closes the gap between formal OpenSCENARIO 2.0 definitions and large-scale naturalistic driving datasets, enabling reproducible and simulation-ready scenario extraction for scenario-based ADAS validation.

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